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Design Report and Bring-Up Guide

Project	Eyecandy Robotics — custom three-phase BLDC motor driver
Schematic file	FOC-driver.kicad_sch (KiCad 10.0.4)
Status	Schematic complete and reviewed; footprints, placement and routing pending
Bus voltage	12 V
Target current	10 A continuous per motor
Intended system	Five of these boards, one per motor, coordinated over a shared communication bus

1. What this board does

1.1 The problem

A **BLDC motor** (Brushless DC motor) is a motor with three sets of coils, called *phases*. Unlike a simple toy motor that spins when you connect a battery, a BLDC motor will not turn at all if you just apply power. It needs an external controller to energise its three coils in a precise, constantly-changing sequence — and the correct sequence depends on exactly where the rotor is at that instant.

So a BLDC motor is only half a system. It has no intelligence of its own. This board supplies the other half.

1.2 What the board provides

The board performs four jobs continuously, thousands of times per second:

1. **Measure where the motor’s rotor currently is** (rotational position).
2. **Measure how much electrical current is flowing** into each of the three coils.
3. **Decide** what each coil should be doing right now.
4. **Deliver** the resulting large currents to the coils.

This closed loop — measure, decide, deliver, measure again — is called **FOC (Field Oriented Control)**. FOC is a mathematically sophisticated control method that produces smooth, quiet, efficient torque at any speed, as opposed to cruder methods that make motors cog and buzz. FOC is *only*

possible if you can measure both position and current accurately, which is why two entire sections of this board are dedicated to measurement.

1.3 Scale of the task

The board switches up to **10 amps** at **12 volts** — roughly 120 watts, comparable to a bright household light bulb, flowing through components the size of a grain of rice. It does so while making decisions on signals measured in **thousandths of a volt**. Keeping the powerful side from disrupting the delicate side is the central engineering challenge, and it explains many decisions in this report.

2. Reading the schematic

A **schematic** is not a picture of the board. It is a *logical map*: it shows which components connect to which, but says nothing about physical position. (Physical position is decided later, during “layout”.) Components are drawn wherever it makes the diagram clearest.

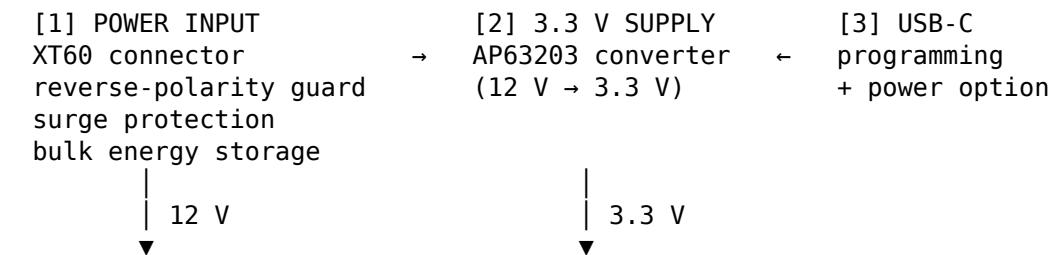
Symbols and conventions used in this schematic:

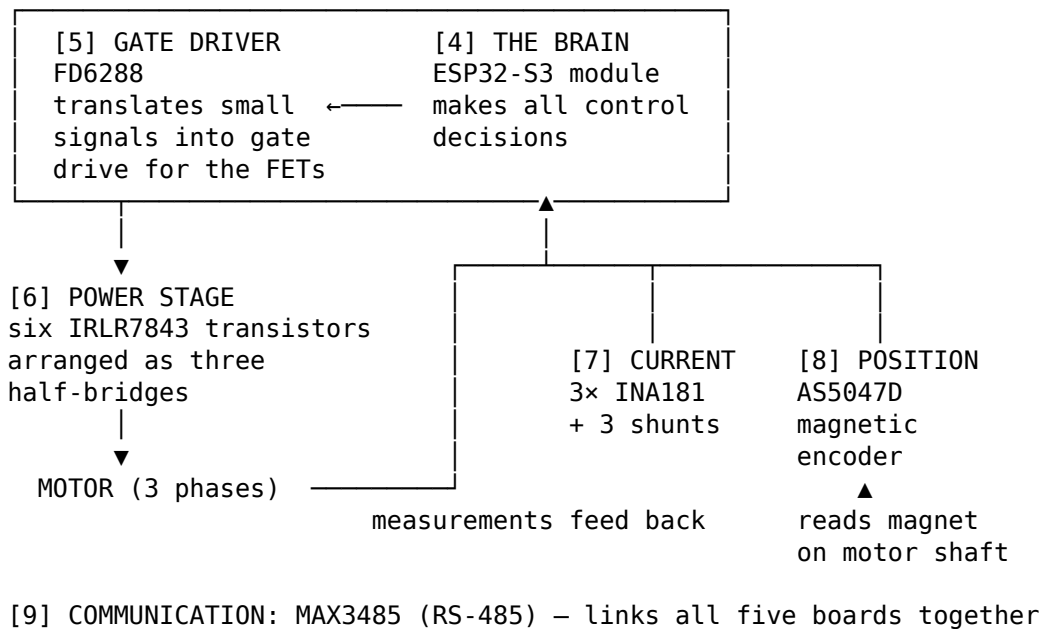
Symbol / label	Meaning
R (e.g. R7)	Resistor — restricts the flow of current
C (e.g. C11)	Capacitor — stores a small amount of electrical charge
L (e.g. L1)	Inductor — a coil that resists sudden changes in current
D (e.g. D1)	Diode — lets current flow one way only
Q (e.g. Q1)	Transistor — an electrically controlled switch
U (e.g. U4)	Integrated circuit — a complete “chip”
J (e.g. J2)	Connector — where wires attach
JP (e.g. JP1)	Jumper — a link that can be fitted or omitted to configure the board
Downward triangle	Ground — the common 0 V reference that all voltages are measured against
10R, 10k, 5mR	10 ohms, 10,000 ohms, 0.005 ohms (resistance values)
0.1uF, 470uF	Capacitor sizes (microfarads)
Named labels (VS1, 12V)	Wires with the same name are connected, even if no line is drawn between them

That last point is important: the schematic uses **named nets** heavily. When you see VB3 in two places, they are the same electrical wire. This keeps the drawing readable instead of a tangle of lines.

3. System overview

The board divides into nine functional blocks. Power flows left to right; information flows in a loop.





4. Block 1 — Power input and protection

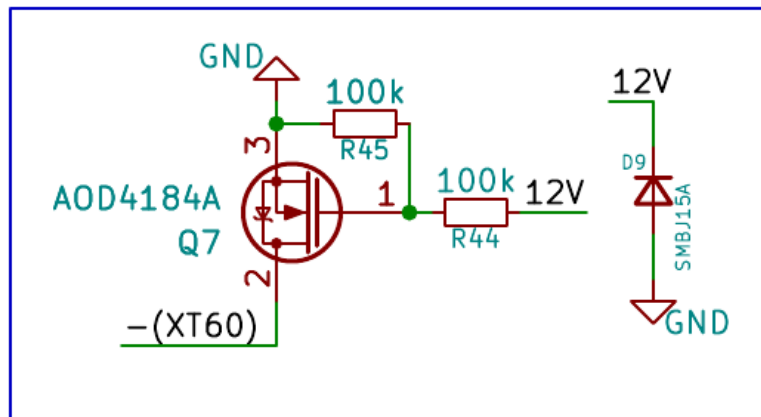


Figure 1: Figure — Reverse-polarity protection (Q7) and surge clamp (D9) on the 12 V input.

Components: J2 / J7 (XT60 connectors), Q7 (AOD4184A transistor), R44, R45, D9 (SMBJ15A), C29 (470 μ F)

This block is the board's front door. Everything that follows depends on it delivering clean, correctly-oriented power.

4.1 The connectors — why two

J2 and J7 are both XT60 connectors. XT60 is a robust, high-current connector widely used in radio-control and robotics; it handles 60 amps, far more than our 10 A, and it cannot be plugged in half-way.

There are *two* of them so that power can be **daisy-chained**: 12 V comes into J2 from the supply, and J7 passes the same 12 V onward to the next board. With five boards in the system, this avoids running

five separate cables back to the power supply.

4.2 Reverse-polarity protection — the most important safety feature

The danger: if the power supply is connected backwards (positive and negative swapped), the board would be destroyed almost instantly. This is not a hypothetical: it is one of the most common ways prototype electronics die, because a connector can be reversed in a moment of inattention.

The reason it is *catastrophic* rather than merely unhelpful is specific to this board. The six power transistors in the motor stage each contain an unavoidable built-in diode (a “body diode”). Under reversed power, all six of these conduct at once, creating a direct short circuit across the power supply through the most expensive components on the board. They vaporise.

The solution — Q7, an electronically controlled gate. Q7 (an AOD4184A transistor) is placed in the negative return path of the power input. It behaves as an automatic gate that only opens when polarity is correct:

- **Correct polarity:** the 12 V rail comes up, which pulls Q7’s control terminal (its “gate”) high through resistor **R44 (100 k Ω)**. Q7 switches fully on, becoming a near-perfect conductor — about 0.007 ohms. At 10 A this wastes only about 0.7 watts, which is negligible. Current flows freely.
- **Reversed polarity:** the control terminal can no longer be driven, so Q7 stays off. Crucially, Q7’s own internal body diode is oriented to *block* in this direction. Both the switch and the diode are closed. **No current flows. The board is protected.**

Why a transistor and not simply a diode? A diode would also block reverse current, and would be far simpler. But a diode always drops about 0.5 V when conducting. At 10 A that is 5 watts of wasted heat, requiring a heatsink, and it steals voltage from the motor. Q7 achieves the same protection with roughly one-seventh the loss. This is the single best reason to accept the extra complexity.

R45 (100 k Ω) connects the control terminal to ground. Its job is to guarantee Q7 is *definitely off* when no power is present, rather than left in an ambiguous half-on state. Without it, the control terminal would be a floating, undefined node during power-up and power-down. This matters especially during bring-up, when the supply voltage is raised slowly (see Section 10).

Why the AOD4184A specifically, and not the same transistor used in the motor stage? Three reasons: it is rated for 40 V rather than 30 V, giving more safety margin against electrical spikes; it has very low resistance (7 m Ω) so it wastes little power at 10 A; and it is built to survive being partially-on for milliseconds, which the motor-stage transistors are not designed for. This last point becomes important if a “soft-start” feature is added later (Section 11.2).

4.3 Surge protection — D9

D9 (SMBJ15A) is a TVS diode, or transient voltage suppressor. It is a safety valve for voltage.

Under normal conditions it does nothing at all — it is effectively an open circuit and electrically invisible. But if the voltage on the 12 V rail suddenly spikes above about 16.7 V, D9 conducts heavily and clamps the voltage to roughly 24 V, absorbing the surge and shunting it to ground.

Why spikes happen here: motors are energy-storing devices. When a motor decelerates it acts briefly as a generator and pushes energy *back* into the 12 V supply, raising its voltage. With five motors in the system, all capable of doing this, voltage spikes are a routine occurrence rather than a rare fault.

Why 15 V was chosen — this was a genuinely tight calculation. The clamping voltage had to be: - **High enough** to stay completely inactive during normal 12 V operation (otherwise it would constantly waste power). - **Low enough** that its ~24 V clamp stays *below* the 25 V rating of the bulk capacitor and the 30 V rating of the motor transistors — the very things it is protecting.

SMBJ15A fits in that narrow window. A higher-voltage version would clamp at 26 V, above the capacitor’s rating, and would therefore fail to protect it.

Placement note: D9 is deliberately connected between the *protected* 12 V rail and the board's ground — that is, on the downstream side of Q7. An earlier draft had it connected to the raw connector terminal, which would have given reverse current a path *around* Q7, defeating the protection. Reverse protection and surge protection must stay in their own lanes.

4.4 Bulk energy storage — C29

C29 is a 470 μF capacitor, by far the largest capacitor on the board. Think of it as a small local reservoir of electrical energy.

Why it is needed: when the motor transistors switch — which happens tens of thousands of times per second — they demand large gulps of current in a few billionths of a second. The power supply and its cables are physically too far away and too “slow” to respond that fast. Without a local reservoir, the 12 V rail would sag and spike violently at every switching event.

C29 sits right next to the demand and supplies those gulps instantly, then refills from the supply between them.

Voltage rating — 25 V, deliberately, and not 16 V. Capacitors of this type should operate at roughly half their rated voltage for long life. On a 12 V rail: - A 16 V part runs at 75 % of its rating — too close, especially given the motor-generated spikes described above. It would run hot and fail early. - A 25 V part runs at 48 % — comfortable margin, cooler operation, long life. **This is the correct choice.** - A 35 V part would be even safer but is physically larger and more expensive without meaningful benefit at 12 V.

5. Block 2 — The 3.3 V supply

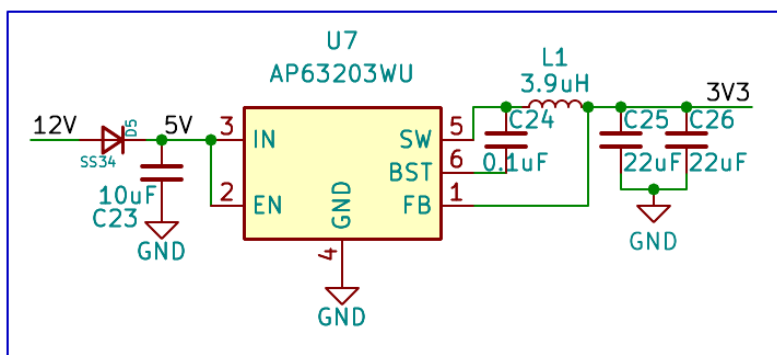


Figure 2: Figure — The AP63203 buck converter stepping 12 V down to 3.3 V.

Components: U7 (AP63203WU), L1 (3.9 μH), C23, C24, C25, C26

The motor runs on 12 V, but every chip on the board — the processor, the sensors, the communication chip — runs on **3.3 V**. This block performs that conversion.

5.1 Why a switching converter

The simplest way to convert 12 V to 3.3 V is a *linear regulator*, which burns the surplus 8.7 V as heat. At the currents this board needs, that would mean wasting more energy as heat than is delivered as useful power — inefficient and hot.

U7 (AP63203) is a switching converter (a “buck converter”) instead. Rather than burning the excess, it chops the 12 V input into rapid pulses and uses an inductor (L1) and capacitors to smooth those pulses into a steady 3.3 V. Efficiency is around 90 % rather than 27 %. It is more complex, but vastly better.

5.2 Why this specific part

The AP63203 was selected against an unusual pair of simultaneous requirements:

1. **It must tolerate 12 V plus spikes.** Its input rating is **32 V** — ample headroom over 12 V and above the surge-clamping level. Many otherwise-suitable converters are rated only 17 V, which is uncomfortably close on a motor bus.
2. **It must also work from about 4.6 V.** This is the non-obvious requirement. When the board is powered from a USB cable instead of the 12 V supply (see Section 6), the input is only 5 V, less a small loss in a protection diode — roughly 4.6 V. The AP63203 operates down to **3.8 V**, so it still produces a clean 3.3 V. Most converters rated for 12 V systems will not go this low, which would mean the board could not be programmed without the main supply connected.

A further benefit, relevant to this board’s noise-sensitive measurements: the AP63203 includes deliberate design features to reduce electrical interference — it slightly randomises its switching frequency and shapes its internal switching edges to avoid generating sharp bursts of noise.

The “203” variant was chosen over the adjustable “200” variant because it has a fixed 3.3 V output built in. The adjustable version would require two additional precision resistors to set the voltage, plus the risk of getting them wrong. Fewer parts, no calculation, no error.

5.3 Supporting components

Part	Value	Purpose
L1	3.9 μ H	The inductor that smooths the chopped voltage. Value taken directly from the manufacturer’s recommendation for 3.3 V output. Verified to be rated for 3.3 A — far above the ~ 1 A this board needs.
C23	10 μ F	Input reservoir. Should be rated ≥ 25 V, as it sees the full 12 V bus.
C25, C26	22 μ F each	Output smoothing, giving a clean, stable 3.3 V.
C24	0.1 μ F	A small internal helper capacitor the chip requires to drive its own internal switch. Without it the converter will not run.

6. Block 3 — USB-C connector

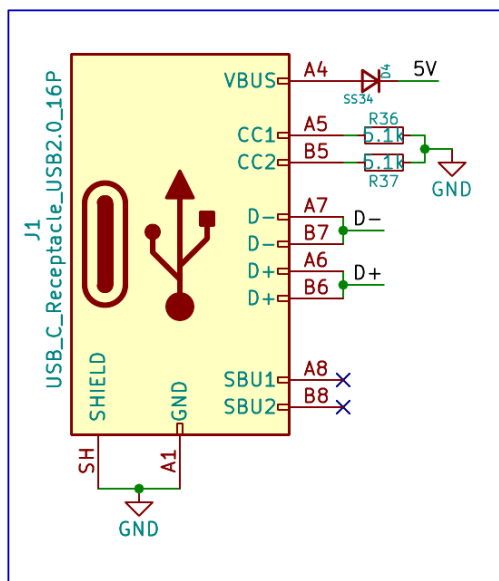


Figure 3: Figure — USB-C receptacle with CC resistors, data-line resistors and the power-share diode.

Components: J1 (USB-C receptacle), R13, R14 (22 Ω), R36, R37 (5.1 k Ω), D4 (SS34), TP1, TP2

USB-C serves two purposes: it carries the programming connection to the processor, and it can optionally power the board.

6.1 Data connection

The two data wires (D+ and D-) run to the processor through **22 Ω resistors (R13, R14)**. These small resistors soften the electrical edges of the high-speed data signals, which reduces radiated interference and improves signal quality — a standard practice for USB.

TP1 and TP2 are test points: small exposed pads that let an engineer attach an oscilloscope probe to observe the data signals during debugging. They cost nothing and are invaluable when something does not work.

6.2 The 5.1 k Ω resistors — not optional

R36 and R37 (5.1 k Ω each) connect to the connector's CC1 and CC2 pins. These tell the USB host, in the language of the USB-C standard, *"I am a device that consumes power, not one that supplies it."*

Without these two resistors, a modern USB-C host will not supply any power at all, and the board will appear completely dead when plugged in. This is one of the most common oversights in first-time USB-C designs, and it is correctly handled here.

6.3 The power-sharing diode — D4

D4 (SS34) is a diode between the USB 5 V supply and the 3.3 V converter's input. Its purpose is to allow either power source to work while keeping them isolated from each other.

- Powered from **12 V**: the converter's input sits at 12 V. D4 is reverse-biased and blocks, so 12 V is **not** pushed back into the USB port — which would violate the USB standard and could damage the connected computer.

- Powered from **USB only**: 5 V passes through D4 (losing about 0.4 V) to give ~4.6 V, which the AP63203 handles comfortably. The board runs, and the processor can be programmed with no main power connected.

This is a small piece of circuitry that removes a real day-to-day annoyance during development.

SBU1 and SBU2 are marked unconnected. These USB-C pins are used only for specialised alternate modes (such as video output) and have no role here.

7. Block 4 — The brain (ESP32-S3)

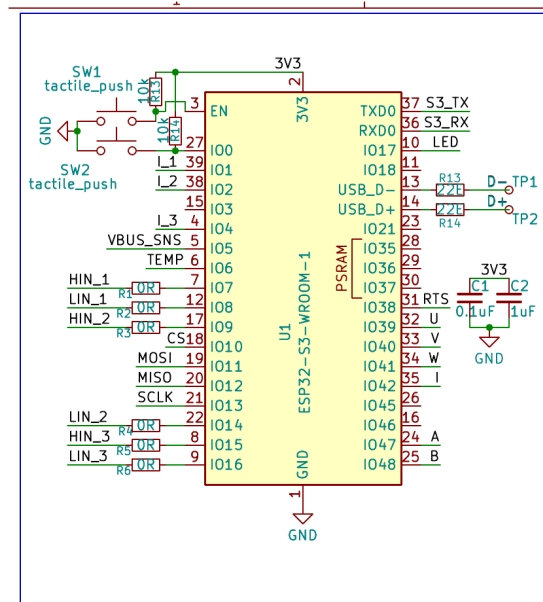


Figure 4: Figure — The ESP32-S3 module — the board’s processor and decision-maker.

Component: U1 (ESP32-S3-WROOM-1 module), SW1, SW2, R13/R14 (10 kΩ pull-ups), D6/D7/D8 (indicator LEDs)

7.1 What it is

The ESP32-S3-WROOM-1 is a self-contained computing module — a processor, memory, and wireless radio on a small board that solders onto ours. It runs the FOC control algorithm: reading position and current, computing what each motor phase should do, and generating the switching commands.

It also provides built-in Wi-Fi and Bluetooth, which is a genuine convenience — control parameters can be tuned wirelessly while the motor is running, without stopping to re-flash software over a cable.

7.2 Why the pin assignments are what they are

This is the most subtle part of the design, and the reasoning is worth explaining because it is invisible on the schematic.

Not all of the module’s pins are freely usable. Some are reserved internally for the module’s own memory. Some are read at power-up to decide how the chip should start, and connecting them wrongly prevents the board from booting. Most importantly:

During the first fraction of a second after power is applied, before any software is running, some pins are briefly driven to a random state by the hardware itself.

For most signals this is harmless. But for the six signals that command the motor transistors, a random pulse at power-up could switch on transistors that must never be on simultaneously — potentially destroying them before the software has even started.

The design deliberately places all six motor-control signals on pins that glitch only to the “off” state, never the “on” state. Combined with two other safeguards (the gate driver’s internal pull-down resistors, and its shoot-through protection described in Section 8.4), this means the board is safe throughout the entire power-up sequence, with no additional components required.

Pins were also kept clear of those reserved for internal memory, of those that control start-up behaviour, and of those reserved for the USB connection and the debug console.

7.3 Series resistors on the control signals

R1 through R6 (0 Ω) sit in series with the six motor-control signals between the processor and the gate driver.

Zero ohms means they are electrically just wires — so why fit them? They are **service points**. Being separate components, any one of them can be removed to isolate a single control signal, or replaced with a resistor of a different value if signal quality needs adjusting after testing. Fitting them costs almost nothing at manufacture; adding them later would require cutting traces on a finished board. This is a mark of a design intended to be debugged rather than merely built.

7.4 Buttons and indicators

- **SW1 and SW2** are the standard **Reset** and **Boot** buttons: one restarts the processor, the other puts it into programming mode. **R13 and R14 (10 k Ω)** hold these inputs in their inactive state until a button is pressed.
 - **D6 (red), D7 (blue), D8 (green)** are indicator LEDs on the 5 V, 3.3 V and 12 V rails respectively. At a glance, an engineer can confirm which power rails are alive — a simple feature that saves considerable time. **R38, R39, R40** limit the current through each LED; they have different values because each rail is a different voltage, and each LED colour requires a slightly different current for equal brightness.
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8. Block 5 — The gate driver (FD6288)

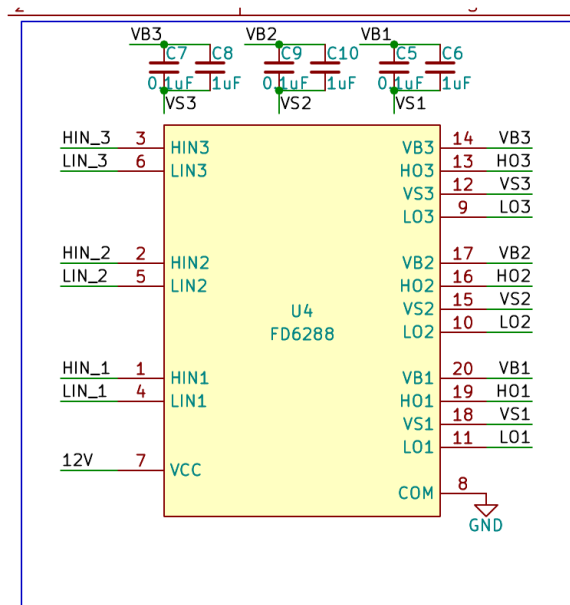


Figure 5: Figure — The FD6288 gate driver translating logic signals into transistor drive.

Components: U4 (FD6288), D1/D2/D3 (SS34 bootstrap diodes), C5–C10 (bootstrap capacitors)

8.1 Why this chip is necessary

It is worth stating clearly why this block is necessary: **the processor cannot drive the motor transistors directly**. Two reasons:

1. **Strength.** The processor's output pins can supply a few thousandths of an amp. Switching a power transistor cleanly requires a brief surge of nearly one full amp — hundreds of times more. Attempt it directly and the transistor switches slowly, overheats, and fails.
2. **Voltage reference — the harder problem.** In each pair of transistors, the upper one has a peculiar requirement: to switch it on, its control terminal must be driven to a voltage *higher than the 12 V supply itself*. The processor only has 3.3 V available. It simply cannot reach.

The FD6288 solves both. It accepts weak 3.3 V logic signals and produces strong, correctly-referenced drive for all six transistors, including generating the above-supply voltage the upper transistors need.

8.2 The bootstrap circuit — how above-supply voltage is created

Each phase has a **diode (D1/D2/D3, type SS34)** and a **pair of capacitors (0.1 μ F and 1 μ F)** forming a *bootstrap* circuit. This is an elegant trick and worth understanding.

1. When the **lower** transistor of a pair is switched on, the connection point between the two transistors is pulled down to 0 V. The diode then allows the capacitor to charge up to nearly 12 V from the supply.
2. When the **upper** transistor is switched on, that connection point rises to 12 V. The charged capacitor rises with it — so its top end is now at roughly 24 V, well above the supply.
3. The gate driver uses that elevated voltage to drive the upper transistor. The diode blocks during this phase, preventing the elevated voltage from flowing back into the 12 V supply.

The capacitor is effectively carried upward on the rising voltage, like a bucket of water lifted while staying full — hence “bootstrap”.

The diode's orientation is critical, and was corrected during design review. The diode must point *into* the capacitor's high side, allowing charge in but not out. Connected the other way round, the upper transistors receive no supply at all and simply never switch on — a failure that is invisible on a schematic and confusing on a bench. All three diodes are now verified correct.

Two capacitors per phase, not one, because they do different jobs: the 1 μF stores enough charge for the whole switching cycle, while the small 0.1 μF responds fast enough to supply the initial instantaneous surge. Neither alone is adequate.

8.3 Power supply for the driver — 12 V, not 3.3 V

The FD6288's supply pin is connected to **12 V**. This is worth stating explicitly, because 3.3 V might seem the natural choice given that the *input signals* are 3.3 V.

The reason: this pin sets the voltage the driver delivers to the transistor gates, and the transistors need roughly 10–12 V on their gates to switch fully on. At 3.3 V they would be only partially on — high resistance, enormous heat, near-certain failure. Additionally, the FD6288 has an internal safety feature that refuses to operate below about 4.6 V, so at 3.3 V it would simply stay shut down.

The 3.3 V logic signals go to the *input* pins; 12 V goes to the *supply* pin. Both are correct, and they are different pins.

8.4 Built-in safety features

The FD6288 was chosen partly for two protections it provides free of charge:

- **Shoot-through prevention.** If both transistors in a pair are ever commanded on at once — through a software bug, electrical noise, or the power-up glitch discussed earlier — they would form a direct short across the 12 V supply and be destroyed in microseconds. The FD6288 detects this contradictory command and switches *both* outputs off instead. It refuses to execute a destructive instruction.
- **Dead time.** Even switching correctly, transistors take a short time to turn off. If the second one turns on too early, there is a brief moment when both conduct. The FD6288 automatically inserts a 200-nanosecond pause between switching one off and the other on, guaranteeing no overlap.

Trade-off, noted honestly: these protections are fixed and cannot be adjusted. The 200 ns dead time is generous, and the chip's internal delays are asymmetric. The practical consequence is that the motor switching frequency should be kept in the **20-40 kHz** range; pushing much higher would waste an increasing fraction of each cycle in the fixed dead time. For this application that range is ideal — above human hearing, so the motor runs quietly, and well within the components' capabilities.

9. Block 6 — The power stage

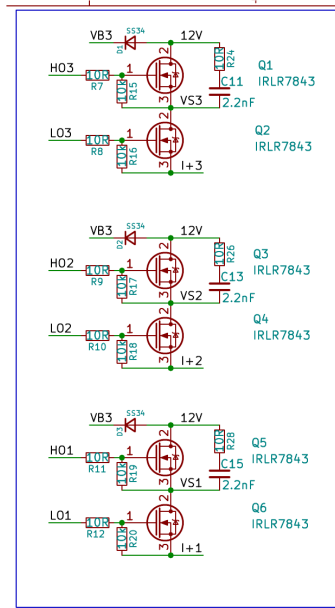


Figure 6: Figure — Three half-bridges (six IRLR7843 transistors) with bootstrap diodes, gate resistors and snubbers.

Components: Q1-Q6 (IRLR7843 transistors), R7-R12 (10 Ω gate resistors), R15-R20 (10 k Ω), R24/R26/R28 with C11/C13/C15 (snubbers)

This is the muscle. Everything else exists to command and monitor this block.

9.1 The arrangement

Six transistors are arranged as **three half-bridges** — three pairs, one pair per motor phase. In each pair, the upper transistor can connect the phase to 12 V and the lower one can connect it to ground. By rapidly alternating, and varying the proportion of time spent in each state, the board can produce any effective voltage between 0 V and 12 V on that phase. Doing this on all three phases in a coordinated pattern is what spins the motor.

The **IRLR7843** was chosen for very low resistance when on (about 3 milliohms), meaning little wasted heat even at 10 A, and for a low control-voltage requirement compatible with the drive available.

9.2 Gate resistors (R7-R12, 10 Ω) — a deliberate compromise

Each transistor's control terminal has a 10 Ω resistor in series. These resistors deliberately **slow down** the switching slightly, and it is worth explaining why anyone would want that.

Without them, the gate driver would deliver current limited only by its own capability — roughly 1.7 A, hitting its internal limit. The transistors would switch as fast as physically possible. That sounds desirable, but:

- Extremely fast switching causes severe electrical “ringing” — violent voltage oscillation (explained in 9.4) that can exceed the transistors' 30 V rating and destroy them.
- Fast edges radiate strong electromagnetic interference, which can disrupt the board's own delicate measurements and cause it to fail electromagnetic-compatibility testing.

The 10 Ω resistors reduce the switching current to about 0.7 A, giving switching times around 20 nanoseconds. This is still extremely fast, but *controlled*. It is a conscious trade of a small amount of efficiency for a large gain in reliability and electrical cleanliness.

9.3 Gate pull-down resistors (R15-R20, 10 kΩ)

Each transistor also has a 10 kΩ resistor from its control terminal to its reference terminal. This guarantees the transistor is held **off** whenever the gate driver is not actively driving it — during power-up, power-down, or a fault.

A transistor's control terminal behaves like a tiny capacitor: left unconnected, it can retain stray charge and drift into a partially-on state, which at 12 V and 10 A means rapid destruction. These resistors ensure the default state is always “off”. They cost fractions of a cent and prevent a whole category of failure.

9.4 Snubbers — why voltage “rings”, and how it is damped

Components: R24/R26/R28 (10 Ω) each paired with C11/C13/C15 (2.2 nF)

The phenomenon. Every time a transistor switches off, the voltage at the motor connection point does not settle cleanly. It overshoots and oscillates rapidly — potentially reaching over 20 V on a 12 V system — before finally settling. This is called *ringing*.

The cause is a natural resonance, and it is worth being precise because it is commonly misunderstood. It is *not* caused by the motor's coils. It is caused by two unavoidable parasitic properties of the board itself:

- The copper tracks and component leads carrying the switching current have a small unavoidable *inductance* — a resistance to sudden changes in current.
- The transistors themselves have a small unavoidable *capacitance*.

Together these form a resonant circuit, much like a bell. Each switching event is a hammer strike, and it rings at tens of millions of cycles per second. This ringing is the single largest source of electromagnetic interference on the board, and its voltage peaks are what threaten the transistors.

The remedy. A resistor and capacitor in series (a *snubber*) placed across the resonance gives that oscillating energy somewhere to go: the capacitor absorbs it, and the resistor converts it to a tiny amount of heat. The ringing dies out in roughly a single cycle instead of dozens.

Note that the gate resistors (9.2) and the snubbers address the *same* problem by different means: the gate resistors strike the bell more gently, while the snubbers damp it once struck. They are complementary, which is why both are present.

9.5 Snubber placement — a subtle and important decision

There are three snubbers, one per phase — not six. An earlier draft had one across every transistor. Two considerations settled this:

1. **Only three are electrically necessary.** Both transistors of a pair share the same connection point, and the 12 V rail behaves as an electrical ground at these very high frequencies. So a single snubber per pair damps the resonance for both. Six would double the parts and the losses for no additional benefit.
2. **The three must be placed on the *upper* transistors, not the lower ones** — and this is the crucial detail. It contradicts the usual textbook advice, for a reason specific to this board.

Placing a snubber across a *lower* transistor would route its absorbed energy through the current-measuring shunt resistor (Section 10). The board would then interpret the snubber's own noise as motor current, corrupting the measurement that FOC depends upon. Placing them across the *upper* transistors routes that energy through the 12 V rail and its capacitors instead, leaving the measurement path completely clean.

This is a good illustration of a general principle: a correct component in the wrong position can silently degrade a system rather than break it outright. Silent degradation is harder to diagnose than outright failure.

Recommendation for assembly: these three snubbers should initially be left **unfitted**. Their optimal values depend on the physical circuit-board layout, which does not exist yet. The correct procedure is to build the board, measure the actual ringing with an oscilloscope, and then calculate and fit the values. There is a reasonable chance they will prove unnecessary if the layout is good — in which case the footprints simply remain empty, at zero cost.

10. Block 7 — Current measurement

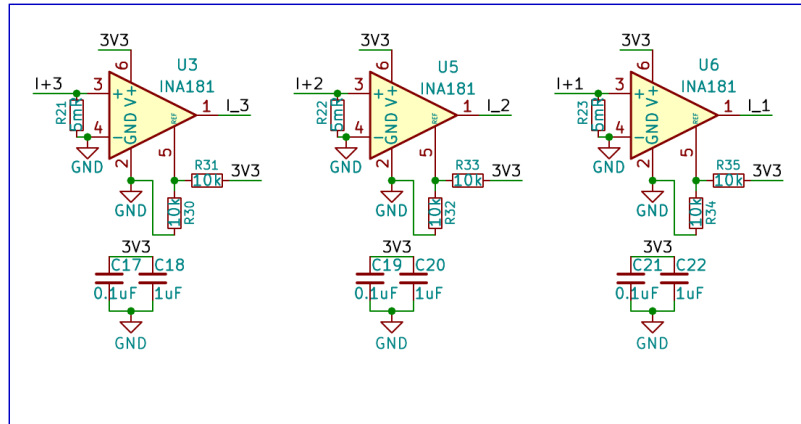


Figure 7: Figure — Three INA181 amplifiers reading the 5 mΩ shunt resistors.

Components: U3, U5, U6 (INA181 amplifiers), R21, R22, R23 (5 mΩ shunts), R30–R35 (10 kΩ), C17–C22

FOC cannot function without knowing the current in each motor phase. This block provides that.

10.1 The shunt resistors

R21, R22, R23 are 5 milliohm resistors — five thousandths of an ohm, essentially a short piece of metal. One sits in the return path of each phase.

The principle is simple: current flowing through any resistance produces a proportional voltage. At 10 A, 5 mΩ produces 0.05 V — a small but measurable signal that is a faithful representation of the current.

The value is a compromise. A larger resistance would give a larger, easier-to-measure signal but waste more power as heat. At 5 mΩ and 10 A, each shunt dissipates about 0.5 watts.

Specification requirement: because of that 0.5 W, these must be specified as **1 watt or higher** power resistors (physically larger than ordinary ones). An ordinary small resistor would overheat and drift, corrupting the measurement. This must be confirmed when parts are ordered.

10.2 The amplifiers

50 millivolts is far too small for the processor to measure reliably, so **U3, U5 and U6 (INA181)** amplify it. These are specialised current-sense amplifiers designed to extract a small signal sitting on top of a large, rapidly-switching voltage — precisely this situation.

10.3 The mid-point reference — enabling negative measurement

R30–R35 (six 10 kΩ resistors, forming three pairs) create a reference voltage of exactly half of 3.3 V, i.e. **1.65 V**, feeding each amplifier's reference pin.

This deserves explanation because it is the difference between a working and a non-working FOC system.

In a motor driven by FOC, current in each phase flows *both directions*, alternating continuously. The processor can only measure voltages between 0 V and 3.3 V — it has no way to represent a negative number directly.

The solution is to shift the zero point. With the reference at 1.65 V: - No current reads as **1.65 V** - Current one way reads **above 1.65 V** - Current the other way reads **below 1.65 V**

The processor simply subtracts 1.65 V in software to recover a signed value. Without this reference the board would measure only one direction of current, and FOC would not work at all.

Specification requirement: the INA181 comes in several amplification strengths. With 5 mΩ shunts and ± 10 A, the **gain-20 variant (INA181A1)** is required. It produces ± 1.0 V around the 1.65 V centre — comfortably within range. The gain-50 variant would attempt ± 2.5 V, exceeding the available range and clipping the measurement at high current. The exact part variant must be specified at ordering.

10.4 Filtering

C17-C22 are small capacitors on each amplifier's supply, providing local clean power and rejecting noise. In a design where an amplifier measuring millivolts sits centimetres from transistors switching 10 A, this local filtering is essential rather than optional.

11. Block 8 — Position measurement (AS5047D)

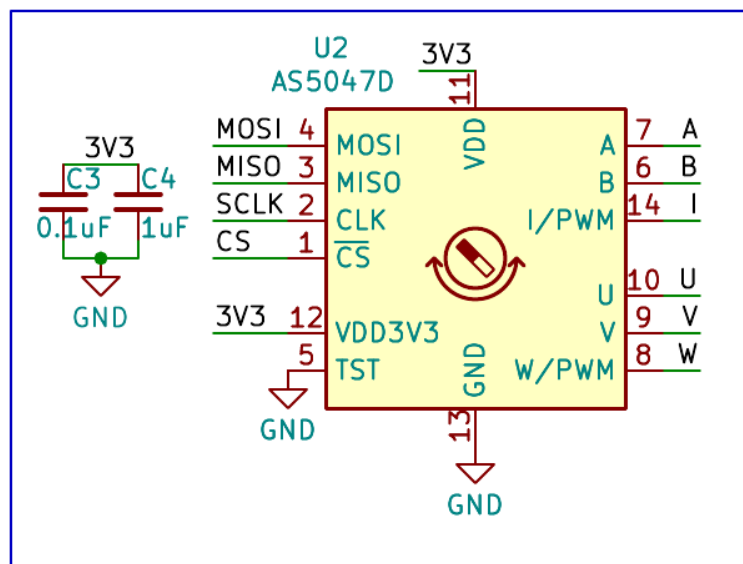


Figure 8: Figure — The AS5047D magnetic encoder on the SPI bus.

Components: U2 (AS5047D), C3, C4

11.1 Why position is indispensable

A BLDC motor's coils must be energised according to where the rotor is *right now*. Get the timing wrong and the motor produces less torque, or vibrates, or turns backwards. Position measurement is not an optional refinement; it is what makes controlled motion possible.

11.2 How it works

The **AS5047D** is a magnetic rotary encoder. A small magnet is fixed to the motor shaft, and the chip sits facing it. As the shaft rotates, the chip senses the direction of the magnetic field and reports the angle.

Its advantages over the alternatives are substantial: - **Non-contact** — nothing touches, so nothing wears out. No maintenance. - **Immune to dirt, oil and dust**, unlike optical encoders. - **Absolute** — it reports the true angle immediately at power-up, without needing to spin the motor first to find a reference point. - **High resolution** — 14 bits, dividing one revolution into 16,384 steps (about 0.02 degrees).

It also includes internal compensation for the small delay inherent in measurement, which matters when the motor is spinning fast: without it, the reported angle lags reality and control quality degrades at speed.

11.3 Connection

The chip communicates with the processor over **SPI**, a four-wire high-speed digital link (CLK, MOSI, MISO, CS). It runs from 3.3 V, matching the processor directly with no conversion needed. **C3 and C4** provide local clean power.

The chip also offers alternative output formats (A/B/I incremental pulses, and U/V/W signals that mimic older sensor types) which are wired out for flexibility, though the SPI link is the primary method.

12. Block 9 — Communication between boards (RS-485)

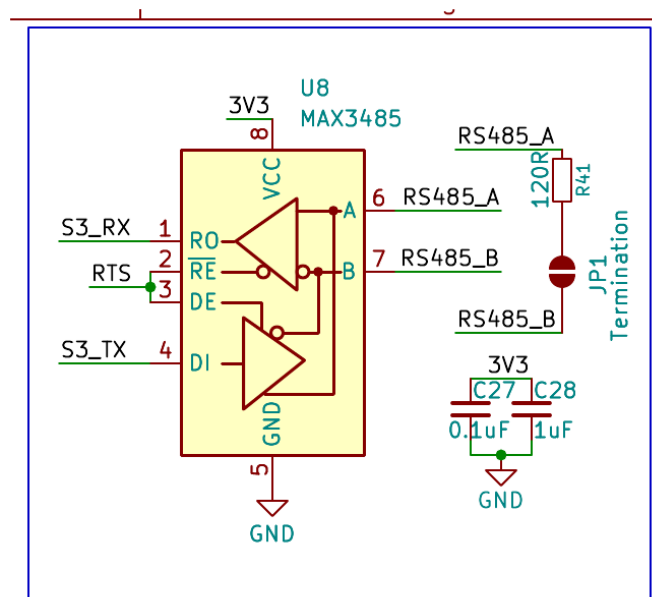


Figure 9: Figure — The MAX3485 transceiver with jumper-selectable termination and bias.

Components: U8 (MAX3485), R41 (120 Ω), JP1, R42/R43 (560 Ω), JP2/JP3, C27, C28

12.1 The requirement

The complete system uses **five of these boards**, one per motor, physically distributed along roughly 1.5 metres of machine. They must be coordinated — receiving target positions and reporting status.

An important architectural clarification, since it is easy to assume otherwise: the **motors themselves are not connected to a shared bus**. Each motor is driven directly by its own board through three short, thick phase wires; those wires must be kept short to avoid the ringing and interference problems discussed in Section 9.4. What runs the 1.5 metres between boards is only the *communication* link — a pair of thin signal wires carrying no significant power.

12.2 Why RS-485

An ordinary digital signal would be unreliable over 1.5 metres in this environment, because five motor inverters generate substantial electrical noise.

RS-485 solves this with differential signalling. Instead of one wire carrying a voltage measured against ground, it uses **two wires (A and B) carrying opposite signals**. The receiver looks only at the *difference* between them. Electrical noise picked up along the cable affects both wires almost identically, so the difference is unchanged — the noise cancels out. This makes RS-485 dramatically more robust than single-wire signalling, and 1.5 metres is well within its capability (it can span tens of metres).

The **MAX3485** was chosen specifically because it operates directly from 3.3 V (matching the processor with no level conversion), is fast enough with wide margin, supports well over five devices, and includes strong built-in protection against static-electricity damage on the exposed cable.

12.3 Direction control — the RTS signal

RS-485 uses the same two wires for both sending and receiving, so each board must be told when to talk and when to listen. The MAX3485's two control pins (RE and DE) are tied together and driven by a single signal, RTS, from the processor: high to transmit, low to receive.

This signal is generated automatically by the processor's hardware, not by software. The processor's serial port has a dedicated mode that raises this signal before the first bit of a message and lowers it immediately after the last. This matters because at high speed, software is not reliably fast enough — a few microseconds of delay would either truncate the end of the board's own message or hold the line occupied into another board's turn to speak. Delegating it to hardware removes that entire class of bug.

A useful side effect: in this mode the chip continues listening while transmitting, so a board can compare what it hears against what it sent. A mismatch means another board spoke simultaneously — free collision detection on a five-node bus.

12.4 Termination — R41 and JP1

R41 is a 120 Ω resistor fitted across the two signal wires, selectable via jumper **JP1**.

At high speeds a cable behaves like a transmission line: a signal reaching an unterminated end **reflects** back down the cable, like an echo, colliding with subsequent data and corrupting it. A 120 Ω resistor matches the cable's natural characteristic impedance and absorbs the signal instead of reflecting it.

Only the two boards at the physical ends of the chain should have this resistor fitted. The three boards in the middle must leave it out — extra terminations would over-load the bus and weaken the signal for everyone. This is exactly why it is on a jumper: all five boards are manufactured identically, and each is configured for its position by fitting or omitting a link. No separate board variants are needed.

12.5 Fail-safe bias — R42, R43 and JP2/JP3

R42 and R43 (560 Ω each) gently pull the A wire toward 3.3 V and the B wire toward ground, again on jumpers.

Their purpose addresses a subtle failure. Between messages, when no board is transmitting, the two wires are left floating at no defined voltage. Receivers then interpret ambient electrical noise as data,

producing streams of meaningless characters that must be filtered out — and which can be mistaken for real messages.

These two resistors hold the idle bus at a small, defined difference that every receiver reads as a clean “nothing happening” state.

This bias must be fitted on exactly one board in the system — hence the jumpers. Five sets in parallel would load the bus excessively. This is a classic issue that works fine when testing two boards on a bench and then fails once the full system is assembled, so it is worth getting right in advance.

13. Summary of key design decisions

Decision	Choice made	Reasoning
Reverse-polarity protection	Transistor, not diode	About one-seventh the power loss at 10 A
Protection transistor	AOD4184A (40 V) rather than reusing the motor transistor	Higher voltage margin; built to survive partial-on operation
Surge clamp level	SMBJ15A (15 V)	Only value that stays inactive at 12 V <i>and</i> clamps below the 25 V capacitor rating
Bulk capacitor rating	25 V, not 16 V	Operating at ~48 % of rating gives long life; 16 V leaves no margin for motor-generated spikes
3.3 V conversion 3.3 V converter part	Switching converter, not linear AP63203	~90 % efficient versus ~27 % Uniquely satisfies both 12 V-plus-spikes input <i>and</i> operation down to 4.6 V for USB-only power
Motor-control pin selection	Pins that glitch “off” at power-up	Makes the board inherently safe during boot with no extra parts
Control signal resistors	0 Ω fitted anyway	Allows post-manufacture isolation and tuning
Gate driver supply	12 V, not 3.3 V	Transistors need ~12 V on their gates; the driver refuses to run below 4.6 V
Gate resistors	10 Ω	Trades slight efficiency for controlled switching, less ringing, and lower interference
Snubber count	Three, not six	Both transistors of a pair share one resonance; six would duplicate losses
Snubber placement	Across upper transistors	Keeps snubber energy out of the current-measurement path
Current-sense reference	1.65 V (half of 3.3 V)	Enables measurement of current in both directions, which FOC requires
Communication	RS-485 differential	Noise immunity across 1.5 m near five motor inverters
Direction control	Hardware-generated	Software timing is not reliable at high data rates
Termination and bias	On jumpers	All five boards identical; configured by position

14. Outstanding items before manufacture

The schematic is electrically complete and reviewed. The following must be resolved during the next stages.

14.1 Immediate — schematic housekeeping

1. **Duplicate component references.** R13 and R14 each appear twice — once as 22 Ω on the USB data lines, and once as 10 k Ω on the buttons. Running KiCad's *Annotate Schematic* function will renumber these automatically. This must be done before generating a parts list, or the wrong values will be ordered.
2. **Run Electrical Rules Check (ERC)** in KiCad to catch any remaining unconnected pins.

14.2 At parts ordering

3. **INA181 variant:** specify the **gain-20 (A1)** version. The gain-50 version will clip at high current.
4. **Shunt resistors:** specify **1 W or higher** for R21/R22/R23. Each dissipates about 0.5 W.
5. **Capacitor voltage ratings:** confirm C29 (470 μ F) and C23 (10 μ F) are rated **25 V or above** — both are exposed to the full 12 V bus and its spikes.
6. **Snubber components** (R24/R26/R28, C11/C13/C15): order but plan to fit *after* measurement. Use low-loss C0G-type capacitors.

14.3 During board layout

These do not appear on the schematic but determine whether the board works:

7. **Heat removal.** All six motor transistors and the protection transistor must have generous copper areas connected to their heat-dissipating tabs, with plated through-holes conducting heat to the opposite side of the board.
8. **Wide copper for high current.** The three motor phase connections and the 12 V and ground paths carry up to 10 A and must be sized accordingly.
9. **Minimise the switching loops.** For each transistor pair, the loop formed by the two transistors and their local capacitors must be as physically small as possible. **This is the single most effective measure against the ringing described in Section 9.4** — more effective than any component that can be added afterwards.
10. **Add local capacitors at each transistor pair.** A 1 μ F and a 0.1 μ F capacitor placed immediately beside each pair (components C30–C35 serve this purpose) supply the instantaneous switching current.
11. **Careful shunt connections.** The connections from each shunt resistor to its amplifier must be taken directly from the resistor's own terminals, not from nearby copper, so that no other current path corrupts the measurement.
12. **Keep the encoder away from switching noise.** The AS5047D and its magnet should be positioned away from the transistors and their copper.

14.4 Noted for the future

13. **Inrush current.** When power is first connected, the 470 μ F capacitor is empty and momentarily draws a very large current, producing a visible spark at the connector and slowly degrading its contacts. The current bring-up plan avoids this by raising the supply voltage gradually from a bench power supply, which is a sound approach. Should a permanent solution be wanted later, adding a single capacitor between Q7's control terminal and ground will make Q7 switch on gradually and eliminate the spark. **No layout change is required** — the AOD4184A was chosen partly because it can withstand this mode of operation.
14. **Grounding across five boards.** Q7 interrupts the negative return path when protection is active. This is correct and safe, but means a board's ground is disconnected from the shared system ground while its power is off. For single-board testing this is irrelevant. For the full five-board system, if all grounds must remain connected regardless of individual power state,

protection on the positive side would be preferable. This is a system-integration consideration, not a fault.

15. Recommended bring-up procedure

Do not connect a motor and apply full power to a new board. A methodical sequence isolates faults while they are still harmless. Each step should pass before the next is attempted.

Step 1 — Visual and continuity check (no power). Inspect for solder bridges. Using a multimeter, confirm there is no short between 12 V and ground, or 3.3 V and ground.

Step 2 — Low-voltage input test. Set a bench power supply to about 5 V with its **current limit set low (~100 mA)**. Connect to J2. If the current limit engages immediately, there is a short — stop and investigate. This step is why a current-limited supply, rather than a battery, is essential for first power-on.

Step 3 — Verify reverse-polarity protection. Still at low voltage and low current limit, deliberately connect the supply **backwards**. Correct behaviour: **no current flows at all**. This confirms Q7 is working, and is far better learned now than by accident later.

Step 4 — Confirm the 3.3 V supply. Return to correct polarity and raise the supply gradually toward 12 V, watching current draw. Measure the 3.3 V rail; it should read between 3.27 V and 3.33 V. The blue and green indicator LEDs should light.

Step 5 — USB and processor. Disconnect the 12 V supply. Connect a USB-C cable to a computer. The board should power up (via D4) and the processor should be detected. Load a simple test program. This validates the USB block, the processor, and USB-only operation together.

Step 6 — Test the gate driver with no motor connected. Reconnect 12 V. With **no motor attached**, run a slow, low-duty-cycle test pattern. Using an oscilloscope, confirm each of the six transistor control terminals switches as commanded, and that the upper and lower transistors of each pair are never on simultaneously.

Step 7 — Verify the bootstrap circuit. Confirm the upper transistors only switch correctly after the lower ones have operated at least once (the bootstrap capacitors must charge first, per Section 8.2). If an upper transistor never switches, suspect its bootstrap diode.

Step 8 — Examine switching quality. Observe the voltage at each motor phase connection with an oscilloscope while switching at the intended frequency. Measure the ringing overshoot. **It must remain safely below 30 V.** If it does not, fit and tune the snubbers (Section 9.5). This is the step at which snubber values are determined.

Step 9 — Calibrate current measurement. With no current flowing, confirm each amplifier outputs approximately 1.65 V. Then pass a known current through each phase and verify the reading scales correctly, including its direction (sign).

Step 10 — Bring up the position encoder. Fit the magnet to the motor shaft. Confirm the AS5047D returns sensible angles over SPI and reports adequate magnet strength through its internal diagnostics. Rotate the shaft by hand and check the angle increases smoothly and does not jump.

Step 11 — First motor rotation. Connect the motor with short phase wires. Set a **low current limit in software**. Attempt slow rotation. Verify the direction matches expectation and that motion is smooth rather than juddering.

Step 12 — Close the control loop and increase gradually. Enable full FOC control. Raise the current limit in stages toward 10 A, monitoring component temperatures at each stage.

Step 13 — Communication. Only once a single board is fully working, connect a second and test RS-485 messaging. Then extend to all five boards, taking care to fit the termination resistors on the two end boards only, and the bias resistors on exactly one board.

16. Glossary

Term	Plain-language meaning
BLDC motor	Brushless DC motor — a motor with three coil sets requiring an external controller to operate
Bootstrap	A circuit trick that generates a voltage higher than the supply, needed to switch the upper transistors
Buck converter	An efficient voltage reducer that chops and smooths rather than burning the surplus as heat
Capacitor	A component that stores a small amount of electrical charge; acts as a local energy reservoir
Dead time	A deliberate brief pause when switching a transistor pair, ensuring the two are never on together
Differential signalling	Sending data as the difference between two wires so that noise, affecting both equally, cancels out
Diode	A one-way valve for electrical current
FOC	Field Oriented Control — a control method producing smooth, efficient, quiet motor torque
Gate	The control terminal of a transistor; the voltage here decides whether it conducts
Gate driver	A chip that converts weak logic signals into the strong, correctly-referenced drive a power transistor needs
Ground	The common 0 V reference against which all other voltages are measured
Half-bridge	A pair of transistors that can connect a point either to the supply or to ground
Inductor	A coil that resists sudden changes in current; used for smoothing
Jumper	A fittable link used to configure a board without redesigning it
MOSFET / transistor	An electrically controlled switch, here handling up to 10 A
Net	A single electrical connection; wires sharing a net name are connected
PWM	Pulse Width Modulation — controlling average power by rapidly switching on and off and varying the proportion
Ringling	Unwanted rapid voltage oscillation after switching, caused by circuit-board resonance
RS-485	A robust industrial communication standard using differential signalling
Schematic	A logical connection map of a circuit; not a physical drawing
Shoot-through	The destructive fault where both transistors of a pair conduct at once, shorting the supply
Shunt	A very low-value resistor used to measure current by the small voltage it develops
Snubber	A resistor-capacitor pair that damps ringling
SPI	A four-wire high-speed digital link between chips
Termination	A resistor at a cable's end that absorbs signals to prevent reflections
TVS diode	A safety valve that conducts only above a set voltage, clamping surges

Term	Plain-language meaning
UVLO	Under-Voltage Lockout — a chip’s refusal to operate below a safe supply voltage

End of report. Schematic reviewed and found electrically sound; proceed to footprint assignment, placement and routing, observing the layout requirements in Section 14.3.